1.44 Hidden Polynomials

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1: Solving Quadratic Equations

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Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

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Test Your Understanding

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1 Solve $(x+3)^2 = x+5$ using factorisation.

2 Solve $(2x+1)^2 = 5$

3 Solve $\sqrt{x+3} = x-3$

4 Solve $2x + \sqrt{x} - 1 = 0$

"Use of Technology":
The 'equation' mode on your calculator will solve quadratics in the form $ax^2+bx+c=0$. When you're asked for the 'order', use 2 (we'll see why later).

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Exercise 2A/2B Pure Y1 Pages 20-22

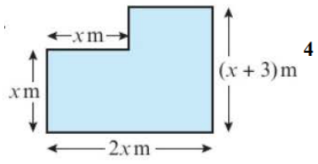
8

3 Solve the following equations:

i $6x^2 - 7 = 11x$

h $5x^2 - 10x^2 = -7 + x + x^2$

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.



4

5 Solve the equation $5x + 3 = \sqrt{3x + 7}$.

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Exercise 2A/2B Pure Y1 Pages 20-22

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2 Solve the following equations using the quadratic formula. Give your answers to three significant figures.

h $2x^2 + 11x = 5x^2 - 18$

3 For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

e $(2x + 8)^2 = 100$

f $6x^2 + 6 = 12x$

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Exercise 2A/2B Pure Y1 Pages 20-22

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Extension: (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 **(i)** Use the substitution $\sqrt{x} = y$ (where $y \geqslant 0$) to find the real root of the equation

$$x + 3\sqrt{x} - \tfrac{1}{2} = 0\,.$$

(ii) Find all real roots of the following equations:

(a) $x + 10\sqrt{x+2} - 22 = 0\,;$

(b) $x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0\,.$

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Test Your Understanding

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1 Find the minimum value of $f(x) = 2x^2 + 12x - 5$ and state the value of x for which this occurs.

2 Find the roots of the function $f(x) = 2x^2 + 3x + 1$

3 Find the roots of the function $f(x) = x^4 - x^2 - 6$

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Quadratics in sin/cos/tan

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We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360^\circ$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

This is the same, but we 'imagine' $\sin x$ as a single variable and hence factorise immediately.

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More Examples

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Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360^\circ$.

Missing the negative case would result in the loss of multiple marks. Beware!

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180^\circ \leq x \leq 180^\circ$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

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Test Your Understanding

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(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360^\circ$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

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Exercise 10F – Y1 P221-222

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5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

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Exercise 10F – Y1 P221-222

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7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

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Exercise 10F – Y1 P221-222

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9\quad $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540^\circ$.

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Quadratics in e^x

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In previous chapters we've already dealt with quadratics in disguise, e.g. "quadratic in sin". We therefore just apply our usual approach: either make a suitable substitution so the equation **is** then quadratic, or (strongly recommended!) **go straight for the factorisation**.

Solve $e^{2x} + 2e^x - 15 = 0$

Note that $e^{2x} = (e^x)^2$ therefore

Solve $e^x - 2e^{-x} = 1$

First write negative powers as fractions:

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Test Your Understanding

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Solve $\ln(3x + 1) = 2$

Solve $e^{2x} + 5e^x = 6$

Solve $2^x e^{x+1} = 3$ giving your answer as an exact value.

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Exercise 14G – Y1 Page 327

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4 $Findtheexactsolutionstothe$ equation $e^{x}+12e^{-x}=7$.

6 Solve $3^{x}e^{4x-1}=5$, giving your answer in the form $\dfrac{a+\ln b}{c+\ln d}$